

An exploration of the rate of convergence of gradient descent

Patrick Huang

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1 Premise

Gradient descent is an iterative process to find the minimum of a function. In recent times, it is extensively used as a numerical method to optimize neural networks, which are functions with a large number of parameters. Because gradient descent and its variants have demonstrated experimental success in the field of machine learning, it can be easy to take for granted that such an algorithm works universally. However, why would we expect such a process to work? And how well does it work?

In this exploration, we will consider gradient descent mathematically, and determine what conditions can be imposed on the function to guarantee convergence, and a rate of convergence.

1.1 Definitions

The function we are trying to optimize is:

$$f : \mathbf{R}^n \rightarrow \mathbf{R}$$

A minimum f^* is the optimal value, and a minimizer x^* is an input that causes it:

$$f^* = f(x^*) \leq f(x)$$

The function is differentiable at all x :

$$\nabla f(x) = \begin{bmatrix} \frac{d}{dx_1} f(x) \\ \frac{d}{dx_2} f(x) \\ \dots \end{bmatrix} \in \mathbf{R}^n$$

The gradient descent process is parameterized by a starting point x_0 and a step size α . The iterative process is defined as:

$$x_{n+1} = x_n - \alpha \nabla f(x_n)$$

1.2 Why gradient?

The gradient reflects the rate of change in an infinitesimally small region around a point. When we use the gradient to decide a macroscopic step, we hope that the gradient carries information about regions beyond the neighborhood of that point.

Using *only* the gradient (and not querying the function at nearby values, for example) simplifies the process and analysis. The search space for a high dimensional function can grow exponentially.

For a differentiable function, we can create a linear model of the local behavior using the first derivative. At any point $x \in \mathbf{R}^n$, taking a small “step” of $v \in \mathbf{R}^n$ results in:

$$f(x + v) \approx f(x) + \nabla f(x) \cdot v$$

Notice that taking a step along the gradient will increase the function, and along the negative gradient decreases the function. The latter is the basis of gradient descent.

$$f(x + \alpha \nabla f(x)) \approx f(x) + \alpha \|\nabla f(x)\|^2 \geq f(x)$$

$$f(x - \alpha \nabla f(x)) \approx f(x) - \alpha \|\nabla f(x)\|^2 \leq f(x)$$

In practice, we want a *macroscopic* step, which is parameterized by the step size α . However, the local linear model of the function may not necessarily apply on a macro scale, so we need to impose additional constraints on the function. The essence of all that follows is how to extend the information in local structures to a global scale.

2 Lipschitz continuity: Somewhat constant gradient

If the gradient is constant, then quite obviously, the gradient at any point gives us information about the function on a global scale. However, such a function would be *globally* linear, which isn't usually a good model.

If keeping the gradient constant doesn't give us any insights, we can instead *limit the rate of change of the gradient*. That is, if the gradient changes, it can only do so gradually; and our local model is valid for some macro scale.

Only with such a constraint can we guarantee anything about the gradient descent process. If the function is free of any constraint, the gradient only carries information about a local, differentiable neighborhood, and we cannot take any macro step with confidence. What if, in the direction of steepest descent, the function suddenly increases? How do we know what a "safe" step size is?

To constrain the gradient, we require that the change in the gradient across two points is bounded by a linear function of the distance:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$$

Although we will not prove it, this is equivalent to the second derivative (the derivative of the gradient) being bounded by $-L$ and L :

$$-LI \preceq \nabla^2 f \preceq LI$$

2.1 Single step, geometrically

Given this constraint, consider what can happen in a single step of gradient descent. For simplicity, first consider a one dimensional input ($f : \mathbf{R} \rightarrow \mathbf{R}$).

Intuitively, this constraint allows the gradient at each point to say something about a macro region around the point. If the function is to violate the linear model, it can only do so sufficiently far away. Geometrically, the function is *upper bounded by a parabola* with second derivative L . Even in the worst case, where the function grows as fast as possible, the growth is limited.

Gradient descent steps in the *direction of steepest descent* — the direction where the function is decreasing locally. Combined with the Lipschitz constraint, there *must* be a macro length before the function can begin increasing again. Imagine we are standing somewhere on a parabola, facing the decreasing direction. It goes down before it can go up.

Let the current input point be x_0 , with output y_0 . Let the derivative at x_0 be a value: $f'(x_0) = m$. The descent direction is toward $-m$. We will derive a mathematical expression for how far we can step while guaranteeing that the function decreases.

In the worst case, $f''(x) = L$, for all x in the macro scale around x_0 (the function increases as fast as allowed). Through algebra, we can derive:

$$f''(x) = L$$

$$\begin{aligned} f'(x) &= \int f''(x) dx \\ &= L(x - x_0) + m \end{aligned}$$

$$\begin{aligned} f(x) &= \int f'(x) dx \\ &= \frac{L}{2}(x - x_0)^2 + m(x - x_0) + y_0 \end{aligned}$$

We can find the minimizer and the set of x where $f(x) \leq f(x_0)$, by cleverly using symmetry around the vertex of the parabola.

$$\begin{aligned} f'(x^*) &= 0 \\ &= L(x^* - x_0) + m \\ x^* &= x_0 - \frac{m}{L} \end{aligned}$$

$$\begin{aligned} f(x_0) &= f(x_0 - \frac{2m}{L}) \\ f(x_s) &\leq f(x_0) \\ x_s &\in [x_0, x_0 - \frac{2m}{L}] \end{aligned}$$

Thus, the largest change in x that guarantees a decrease in the function is $\Delta x = \frac{2m}{L}$. Because our step is the product of the gradient and step size, we have:

$$\begin{aligned} \Delta x &= \frac{2m}{L} \\ &= \alpha \nabla f(x) \\ \nabla f(x) &= m \\ \alpha &= \frac{2}{L} \end{aligned}$$

Consider that last statement. In gradient descent, the magnitude of the step taken is determined by *both* the step size parameter (α) and the magnitude of the gradient (∇f). The “step size” α is actually a *step scaling*.

2.2 Single step, algebraically

Because the second derivative of f is constrained, we know that f is upper bounded by a quadratic on a global scale.

$$f(x) \leq f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{L}{2} \|x - x_0\|^2$$

That is, at any point x_0 , construct a quadratic. Match the value and the first derivative to that of the function. Set the second derivative be the maximum possible, L . The function must be below this globally.

Now consider a gradient descent step, and the difference in function values across the step.

$$x_1 = x_0 - \alpha \nabla f(x_0)$$

$$\begin{aligned} f(x_1) - f(x_0) &\leq \nabla f(x_0) \cdot (-\alpha \nabla f(x_0)) + \frac{L}{2} \|\alpha \nabla f(x_0)\|^2 \\ &= -\alpha \|\nabla f(x_0)\|^2 + \frac{L}{2} \alpha^2 \|\nabla f(x_0)\|^2 \\ &= \alpha \left(\frac{1}{2} L \alpha - 1 \right) \|\nabla f(x_0)\|^2 \end{aligned}$$

If the gradient descent step is to decrease the function value ($f(x_1) - f(x_0) \leq 0$), we can derive a condition for α :

$$\begin{aligned} \alpha \left(\frac{1}{2} L \alpha - 1 \right) \|\nabla f(x_0)\|^2 &\leq 0 \\ \alpha \left(\frac{1}{2} L \alpha - 1 \right) &\leq 0 \\ \alpha &\in \left(0, \frac{2}{L} \right) \end{aligned}$$

2.3 The entire process

Now we are going to “reverse the implication” of the above inequality. Previously, choosing a step size guarantees that the function decreases after a single iteration. Here, we will bound the amount the function can decrease to *show that the gradient must eventually disappear*.

Consider gradient descent for multiple steps, from 0 to K . We already have a (negative) upper bound for the change in a single iteration. Summing this across the entire process produces an overall upper bound:

$$\begin{aligned} C &= \frac{1}{2} L \alpha^2 - \alpha \leq 0 \\ 0 &\geq C \|\nabla f(x_i)\|^2 \geq f(x_{i+1}) - f(x_i) \\ 0 &\geq \sum_{i=0}^{K-1} C \|\nabla f(x_i)\|^2 \geq \sum_{i=0}^{K-1} f(x_{i+1}) - f(x_i) \\ &= f(x_K) - f(x_0) \end{aligned}$$

Additionally, we can derive an overall lower bound as well, by reasoning about the nature of a function. If we assume f has a minimizer x^* , such that $f(x^*) \leq f(x)$ for all x , then

$$f(x^*) \leq f(x_K) \leq f(x_0)$$

$$f(x^*) - f(x_0) \leq f(x_K) - f(x_0)$$

Putting the upper and lower bounds together, we get:

$$\begin{aligned} 0 &\geq \sum_{i=0}^{K-1} C \|\nabla f(x_i)\|^2 \\ &\geq f(x_K) - f(x_0) \\ &\geq f(x^*) - f(x_0) \end{aligned}$$

The first and last terms of this inequality are *constants* that don’t depend on K — independent of how long we run the process.

Consider what this means for the summation term. Because C is negative and the magnitude of the gradient is positive, this is a summation of negative numbers. As we take $K \rightarrow \infty$, the series remains bounded. This means that the values themselves necessarily tend to zero.

$$\begin{aligned}
0 &\geq C \|\nabla f(x_i)\|^2 \\
0 &\geq \lim_{K \rightarrow \infty} \sum^K C \|\nabla f(x_i)\|^2 \\
&\geq f(x^*) - f(x_0) \\
\implies 0 &= \lim_{K \rightarrow \infty} C \|\nabla f(x_i)\|^2
\end{aligned}$$

This means that in the limit, we reach a point where the gradient becomes zero. If $\nabla f(x_i) = 0$, then $x_{i+1} = x_i$, so the process is guaranteed to converge. Not necessarily to a minimum, however.

2.4 Summary

For the gradient to say anything about the function on a macro scale, we constrained the second derivative, so the gradient stays *somewhat* constant.

This necessarily implies that there is some macro step size at each point that *guarantees* the function to decrease. If we step toward the direction of steepest descent, the function can't instantaneously ascend.

We choose such a step size, and run the gradient descent process. The function is decreasing at every step, but the *total amount of decrease is limited*, if we assume that the function has a minimizer.

Therefore, in the limit, the steps must become smaller and smaller, and tend to zero. The gradient descent process is *guaranteed to converge*, where the gradient becomes zero, and the function value is a (possibly local) minimum.

3 Strong convexity: Predictable convergence rate

The Lipschitz constraint in the previous section is a quadratic upper bound on the function value. Here, we will explore “strong convexity”, which forms a lower bound.

Convex functions are functions that can only “curve up”. One characterization is that the second derivative is nonnegative everywhere.

The second derivative of a *strongly* convex function is not only positive, but greater than some positive threshold everywhere. They *must curve up at least some amount*. Another characterization of convexity is the tangent. A strongly convex function lies above the tangent parabola at every point.

The previous Lipschitz constraint allowed us to choose a step size that is guaranteed to converge. Strong convexity guarantees an exponential rate of convergence.

3.1 Gradient descent on a parabola

Because the essence of strong convexity is a quadratic lower bound, let's first consider gradient descent on a parabola.

Let the function be:

$$\begin{aligned}
f(x) &= \frac{1}{2} M x^2 \\
M &> 0
\end{aligned}$$

Quite obviously, the optimum is $x = f'(x) = 0$.

Choose a step size. The second derivative is $f''(x) = M$, so the gradient is M Lipschitz continuous. Therefore, any $\alpha < \frac{2}{M}$ guarantees that the descent process decreases at every iteration. Intuitively, the step size is small enough that the process doesn't “jump” far across the parabola's vertex, to a point where function value is higher.

Consider a starting point $x_0 > 0$. By gradient descent, we have:

$$\Delta x = x_{n+1} - x_n = -\alpha f'(x_n) = -\alpha M x_n$$

$$\Delta x_n \propto -x_n$$

The key insight is that the step magnitude is proportional to the suboptimality of the input. Because step magnitude is proportional to gradient, and gradient is proportional to x for a parabola, we get this relation.

Solving the difference equation, we obtain (roughly) the following. Gradient descent on a parabola converges to the optimum at *an exponential rate*.

$$x \propto e^{-n}$$

Finally, because function value is proportional to the square of the input, we have:

$$f(x) \propto x^2 \propto e^{-2n}$$

Because of the shape of a parabola, the succeeding value $x_{n+1} \in (-x_n, x_n)$ guarantees a decrease of function value. However, the process may either oscillate or monotonically converge. We explore the different behaviors in section 5.

3.2 Gradient domination: A proportional gradient

Now, let's mathematically consider the convergence rate. We will show that strong convexity implies an intermediate conclusion, the *gradient domination condition*, which then implies an exponential convergence rate.

A function is strongly convex with parameter μ if the function lies above every tangent parabola. That is, for every point on the function, fit a parabola by matching the value and first derivative, and setting the second derivative equal to μ .

$$f(y) \geq f(x) + \nabla f(x) \cdot (y - x) + \frac{1}{2}\mu\|y - x\|^2$$

To show the gradient domination condition, take the min of both sides of the inequality with respect to y . The LHS is simply the minimum value of f . The RHS is a quadratic with respect to y , so we can find the minimum by setting the derivative equal to zero.

$$\begin{aligned} 0 &= \frac{d}{dy}(f(x) + \nabla f(x) \cdot (y - x) + \frac{1}{2}\mu\|y - x\|^2) \\ &= \nabla f(x) + \mu(y - x) \\ \implies y &= x - \frac{1}{\mu}\nabla f(x) \end{aligned}$$

$$\begin{aligned} \min(f(y)) &\geq \min(f(x) + \nabla f(x) \cdot (y - x) + \frac{1}{2}\mu\|y - x\|^2) \\ f^* &\geq f(x) + \nabla f(x) \cdot \left(-\frac{1}{\mu}\nabla f(x)\right) + \frac{1}{2}\mu\left\|-\frac{1}{\mu}\nabla f(x)\right\|^2 \\ f^* - f(x) &\geq \|\nabla f(x)\|^2 \left(-\frac{1}{2\mu}\right) \\ f(x) - f^* &\leq \frac{1}{2\mu}\|\nabla f(x)\|^2 \end{aligned}$$

The last inequality is the gradient domination condition. The magnitude of the gradient is larger than the function suboptimality, up to a scale factor.

Intuitively, this means: Take an arbitrary point $(x, f(x))$. The function minimum, f^* , is lower than $f(x)$ *somewhere* in the domain. Further, by strict convexity, we know that a tangent parabola at $(x, f(x))$ must lie below the entire function — including f^* . The tangent parabola at this point has a fixed 2nd derivative (μ) and value ($f(x)$). Therefore, it's derivative, $\nabla f(x)$, must be *large enough* such that the parabola can decrease below f^* .

We have an inequality for a single step of gradient descent, obtained with the Lipschitz condition (section 2.2). Now, we can combine it with the gradient domination condition. With some algebra, we obtain the subsequent suboptimality as a fraction of the preceding.

$$\begin{aligned}
f(x_1) - f(x_0) &\leq \alpha \left(\frac{1}{2} L \alpha - 1 \right) \|\nabla f(x_0)\|^2 \\
f(x_1) - f^* &\leq f(x_0) - f^* + \alpha \left(\frac{1}{2} L \alpha - 1 \right) \|\nabla f(x_0)\|^2 \\
&\leq f(x_0) - f^* + \alpha \left(\frac{1}{2} L \alpha - 1 \right) (2\mu) (f(x_0) - f^*) \\
&= (f(x_0) - f^*) (1 + \mu \alpha (L \alpha - 2)) \\
&= (f(x_0) - f^*) \rho
\end{aligned}$$

$$f(x_n) - f^* \propto e^{-n}$$

The sequence converges if the factor is less than one. Because both μ and α are positive, we obtain the same condition as the previous section.

$$\begin{aligned}
\rho &= 1 + \mu \alpha (L \alpha - 2) < 1 \\
1 + L \mu \alpha \left(\alpha - \frac{2}{L} \right) &< 1 \\
L \mu \alpha \left(\alpha - \frac{2}{L} \right) &< 0 \\
\alpha &< \frac{2}{L}
\end{aligned}$$

If there was no change on the step size constraint, what intuitively happened when we introduced the gradient domination condition? In a sense, we guaranteed that the process is exponential. Step magnitude is determined by gradient magnitude, which must be larger than the suboptimality. Therefore, when the point is far from the minimum, the step must be large.

When is the convergence fastest? The expression ρ contains a positive quadratic in α ; by inspection, the minimum is at $\alpha = \frac{1}{L}$.

$$\begin{aligned}
\arg \min (1 + L \mu \alpha (\alpha - \frac{2}{L})) &= \frac{1}{L} \\
\rho &= 1 - \frac{\mu}{L}
\end{aligned}$$

Now, the convergence rate ρ is dependent on the disparity between μ and L . The Lipschitz L is a quadratic upper bound, while the strong convexity μ is a quadratic lower bound. Clearly, $\mu \leq L$. The closer they are, the faster the process converges. If $\mu = L$, the function *is* a quadratic, and choosing $\alpha = \frac{1}{L}$ converges in *a single step*. We will explore these in sections 4 and 5.

Additionally, every minimizer of a convex function is a global minimizer. Therefore, if the process finds a minimum, it is the global minimum of the function.

3.3 Summary

In the previous section, we upper bound the function, which guarantees that the gradient descent process converges.

Here, we lower bound the gradient magnitude with the gradient domination condition, as a function of suboptimality. Therefore, the function value decreases by some nonzero amount each iteration, proportional to the suboptimality. This creates an exponential convergence rate.

4 Eigenvalue disparity

Consider the function and its Hessian:

$$f(x, y) = x^2 + 0.01y^2$$
$$\nabla^2 f = \begin{bmatrix} 2 & 0 \\ 0 & 0.02 \end{bmatrix}$$

By observation, the minimizer is $(0, 0)$.

If we run gradient descent process on this function from an arbitrary starting point, we would notice something interesting. The x component converges *much faster* than the y component. The x approaches 0 rapidly, while the y slowly decreases. It's still exponentially decays — just slowly.

4.1 Upper and lower quadratic bound

We have used a quadratic bound in two ways: The Lipschitz continuity of the gradient is an upper quadratic bound on the function value. The strong convexity parameter is a lower bound. As we saw above, the convergence rate depends on both of these constants.

Crucially, the Lipschitz L determines the step size, while the strong convexity μ determines convergence rate. We choose the step size α , and our choice is *based on the gradient Lipschitz*. This is because the upper bound defines the worst case — the fastest that the function can grow. This limits the largest step size we can choose.

On the other hand, the “guaranteed convergence rate” is defined by the strong convexity constant. Clearly, this must be lower than the Lipschitz constant (lower bound is lower than upper bound). The closer this is to the Lipschitz constant, the faster the process will converge, as we saw in the previous section.

The above example is a two dimensional function, which has a different second derivative in the x and y direction. The Lipschitz value is the maximum of these, which is 2. This determines the max step size. A larger step size might speed up convergence in the y direction, but we cannot choose it confidently; for that would risk the process exploding in the x direction.

5 Dynamical system around the minimizer

Consider gradient descent as a discrete dynamical system. Here, “dynamical system” means that we track a state x_k , and the next state is defined by a function $x_{k+1} = g(x_k)$.

Let f be the function we are trying to optimize, and let g be *the result of a single step of gradient descent*.

$$g(x) = x - \alpha \nabla f(x)$$

Assuming f is differentiable, the derivative at a minimizer must be zero. Therefore, a minimizer is a fixed point of g :

$$\begin{aligned} \nabla f(x^*) &= 0 \\ g(x^*) &= x^* \end{aligned}$$

5.1 Example: Parabola

Let $f(x) = x^2$. The function both has Lipschitz gradient and is strongly convex, with the same parameter. Therefore, we expect an exponential convergence rate.

We get g as a linear function (because ∇f is linear):

$$\begin{aligned} g(x) &= x - \alpha(2x) \\ &= x(1 - 2\alpha) \end{aligned}$$

The unique fixed point of g is the minimizer $x^* = 0$. Using cobweb plots or analysis, we see a few distinct regions of behavior depending on α .

$\alpha = 0$: The function $g(x) = x$ becomes the no-op. An initial point x_0 stays where it is ($\rho = 1$).

$\alpha \in (0, 0.5)$: All initial points approach 0 at an exponential rate ($\rho < 1$).

$\alpha = 0.5$: All points reach 0 after *a single iteration* ($\alpha = \frac{1}{L}$, $\rho = 0$).

$\alpha \in (0.5, 1)$: All initial points exponentially approach 0, but oscillating (changing sign every iteration) ($\rho < 1$).

$\alpha = 1$: All initial points oscillate stably, maintaining magnitude and changing sign ($\alpha = \frac{2}{L}$, $\rho = 1$).

$\alpha > 1$: Points explode away from 0 exponentially, while oscillating ($\rho > 1$).

And just for completeness, $\alpha < 0$: Points explode away without oscillating. This is gradient *ascent*.

5.2 Stability

Now let's consider the dynamical system more generally.

A condition of stability on dynamical systems is that the map function, g , be 1-Lipschitz continuous around a fixed point. That is, for a fixed point x^* :

$$\begin{aligned}\|g(x) - g(x^*)\| &< \|x - x^*\| \\ \equiv \|g(x) - x^*\| &< \|x - x^*\|\end{aligned}$$

Intuitively, this means that applying the function g must take the orbit x *closer* to x^* than it previously was. A rigorous statement requires conditions on the continuity of g , but for a well behaved function that satisfies the above, orbits converge to x^* .

First, consider a 1D case ($x \in \mathbf{R}$):

$$\begin{aligned}x &> x^* \\ g(x) - x^* &\in (x^* - x, x - x^*) \\ g(x) = x - \alpha \nabla f(x) &\in (2x^* - x, x) \\ -\alpha \nabla f(x) &\in (2x^* - 2x, 0)\end{aligned}$$

$$0 < \nabla f(x) < \frac{2}{\alpha}(x^* - x)$$

This is the Lipschitz gradient condition. The gradient descent process converges if the rate of change of the gradient is bounded. Note that $\nabla f(x^*) = 0$, a necessary condition for optimality.